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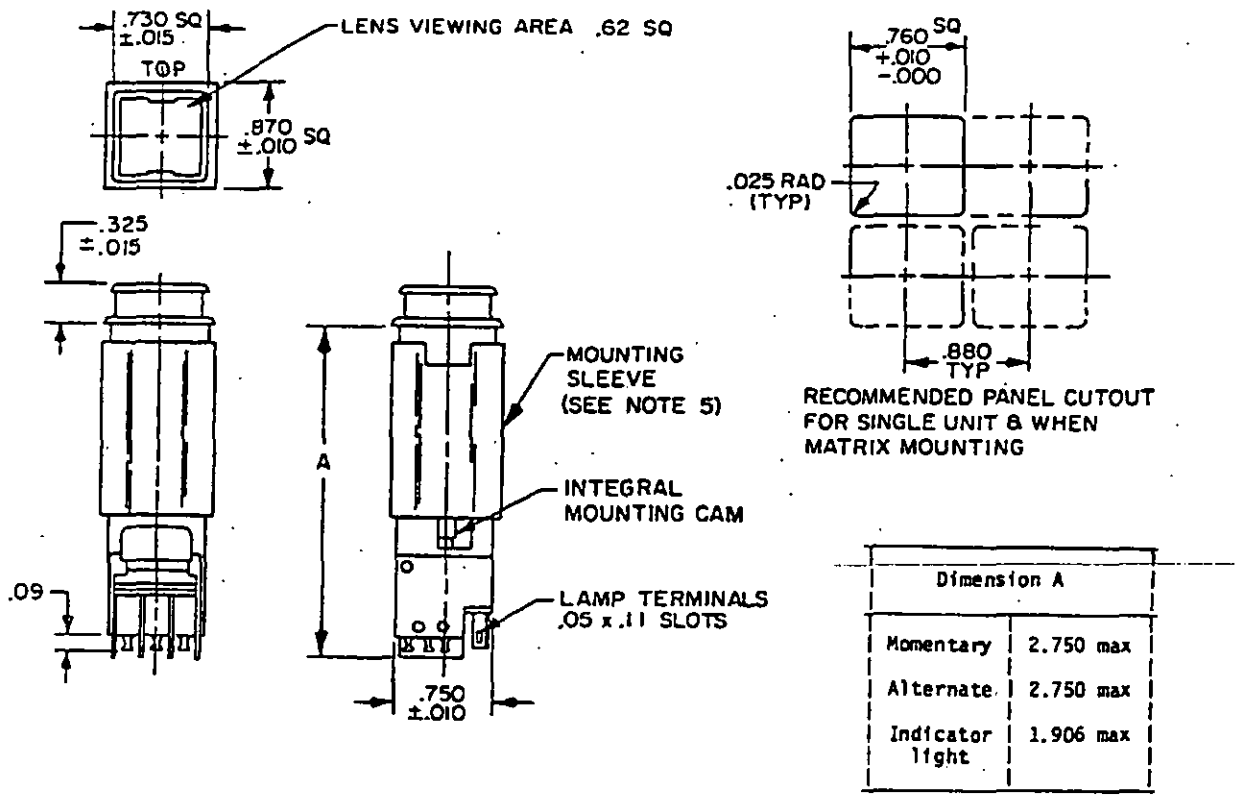
INCH-POUND
MIL-PRF-22885/75D
25 July 1990
SUPERSEDING
MIL-S-22885/75C
18 December 1987

PERFORMANCE SPECIFICATION SHEET

SWITCH, PUSH BUTTON, ILLUMINATED, 4-LAMP, 0.87 SQUARE, 7 AMPERES, ELECTROMAGNETIC INTERFERENCE SHIELDED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-22885.



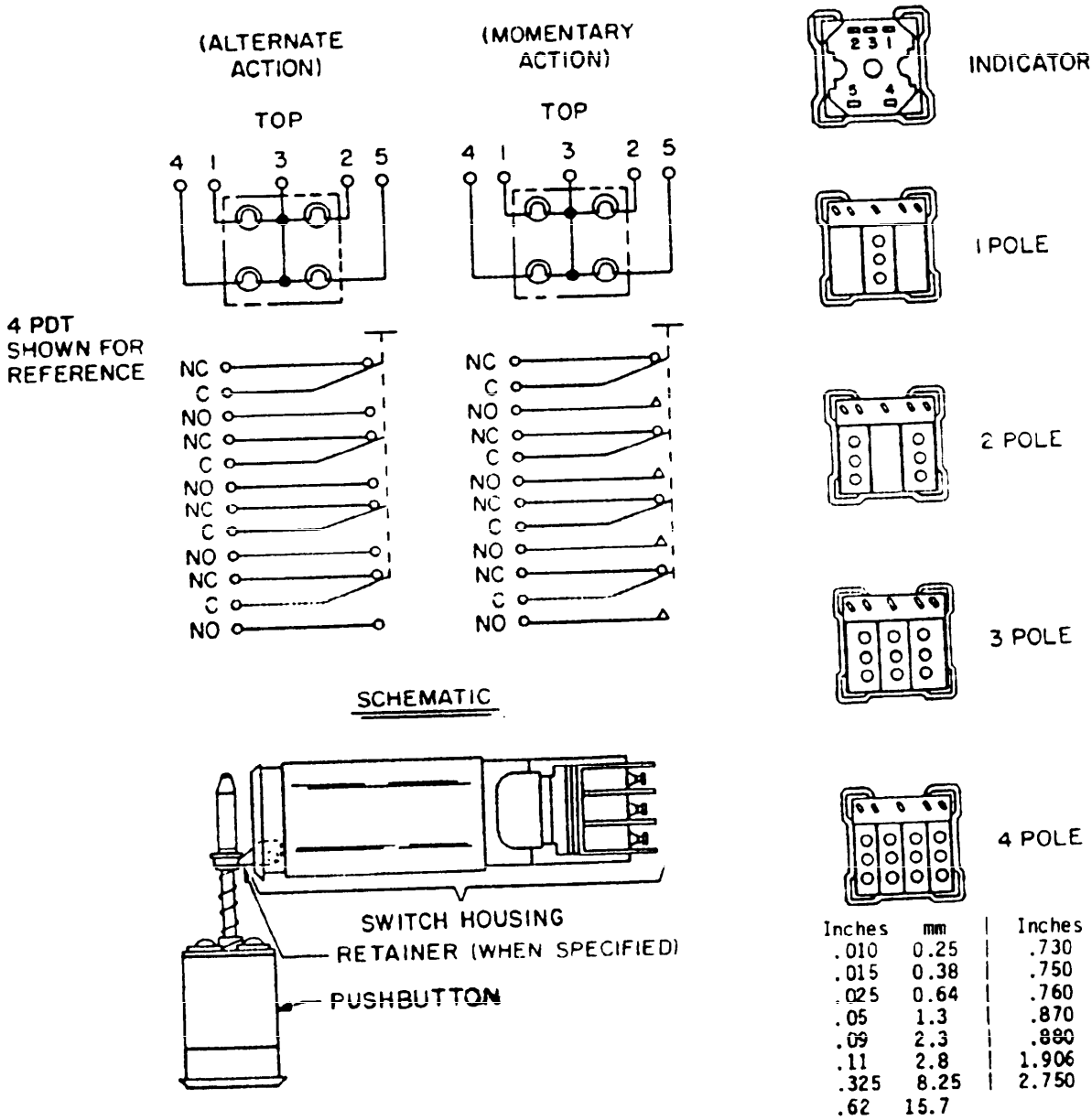

| Dimension A | |
|---|---|
| Momentary | 2.750 max |
| Alternate | 2.750 max |
| Indicator light | 1.906 max |

FIGURE 1. Design and construction.

(D) Denotes changes


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| Inches | mm | Inches | mm |
|---|---|---|---|
| .010 | 0.25 | .730 | 18.54 |
| .015 | 0.38 | .750 | 19.05 |
| .025 | 0.64 | .760 | 19.30 |
| .05 | 1.3 | .870 | 22.10 |
| .09 | 2.3 | .880 | 22.35 |
| .11 | 2.8 | 1.906 | 48.41 |
| .325 | 8.25 | 2.750 | 69.85 |
| .62 | 15.7 | | |

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are ±.005 (0.13 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.
3. Metric equivalents are given for general information only.
4. Unit mounts in panels from .125 (3.18 mm) to .250 (6.35 mm) thick.
5. Mounting sleeve shall not exceed .87 square and shall be reversible to accommodate increased panel thickness.
6. Pushbutton designed to prevent incorrect insertion into switch housing.
7. Exact shape of switch optional provided dimensions specified are not exceeded.

FIGURE 1. Design and construction - Continued.



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REQUIREMENTS:

Design and construction: See figure 1. Basic switches shall be MIL-S-8805/4 (MS24547-1) or MIL-S-8805/109 (M8805/109-01) listed in QPL-8805.

Enclosure design: 1 (unsealed).

Temperature characteristic: 1 (-55°C to +85°C).

Vibration grade: 2 (10-500 Hz).

Operation: A (momentary action), B (alternate action), H (indicator light).

Illuminated color: See table I.

Display type:

W (white until illuminated, then background appears in color).

Material:

Housing: Corrosion resistant steel or aluminum alloy.

Ⓓ Weight: 2.2 ounces maximum.

Lamps: Pushbutton shall accommodate four lamps, T-1 3/4 midget flange base. Lamps shall be acquired separately.

Operating characteristics:

Actuation force: 2 to 5 pounds.

Ⓓ Pushbutton travel: .218 ±.031.

Lens extraction force: 2 to 8 pounds.

Shock: MIL-STD-202, method 213, test condition H, (75 G's for 6 milliseconds, sawtooth).

Moisture resistance: Following step 6 of the final cycle of the moisture resistance test, while switches are still wet, the insulation resistance measurement shall be not less than 5 megohms.

Electrical endurance: See table II for electrical load conditions.



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Ⓓ TABLE I. Illuminated color. 1/

| Chromaticity coordinates 1/ 2200° kelvin | | |
|---|---|---|
| Colors | X | Y |
| Green | .350 | SL 2/ |
| | .350 | .610 |
| | .185 | .610 |
| | .270 | SL 2/ |
| Amber | .600 | SL 2/ |
| | .588 | .385 |
| | .627 | .346 |
| | .640 | SL 2/ |
| Yellow | .530 | .442 |
| | .544 | SL 2/ |
| | .571 | .402 |
| | .584 | SL 2/ |
| Red | 680 | .300 |
| | .690 | .310 |
| | .723 | .257 |
| | .736 | .264 |
| Blue | .180 | .300 |
| | .180 | .370 |
| | .120 | .370 |
| | .120 | .300 |
| White | .484 ref | .415 ref |
| X is not less than .350. Y is not greater than .520. $Y - Y_o$ or $Y_o - Y$ is not numerically greater than 0.05X. ($Y_o$ is the Y coordinate of the Planckian radiator for a given X). | | |

1/ The chromaticities of the diffusing lenses or colored filters expressed as X and Y coordinates on the CIE chromaticity diagram shall lie within areas bounded by the coordinates listed for each color, using MS25237-387 lamps .30 ±.01 mean spherical candlepower at 28 ±.02 V dc reference 2075°/2100° kelvin.

2/ SL = spectrum locus (where intersected by other of coordinate pair).



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TABLE II. Electrical ratings.

| Load | Sea level | | 50,000 feet 28 V dc |
|---|---|---|---|
| | 28 V dc | 115 V ac, 60 Hz | |
| | (Amperes) | (Amperes) | (Amperes) |
| Resistive | 7 | 7 | 4 |
| Inductive | 4 | 7 | 2.5 |
| Lamp | 2.5 | 2 | 2.5 |

EMI/RFI shielding. Shielding efficiency test applicable, except the shielding attenuation shall be not less than 47 dB at 100 MHz, 39 dB at 400 MHz and 33 dB at 1,000 MHz.

Luminance: See table III.

TABLE III. Luminance.

| Color | Color code | Brightness 1/ 2/ (foot-lamberts, minimum average) |
|---|---|---|
| Blue | B | 12.0 |
| Green | G | 80.0 |
| Red | R | 56.0 |
| White | W | 146.0 |
| Amber | A | 406.0 |
| Yellow | Y | 470.0 |

1/ When illuminated by two MS25237-327 lamps of 0.34 ±0.02 mean spherical candlepower at 28 volts dc, readings are for lens configuration 0.

2/ Brightness is reduced by approximately 50 percent when EMI screen is specified.

Marking: Lamp circuit schematic shall be marked on the switch housing.

Qualification inspection:

Group submission: All part numbers will be approved for inclusion in the qualified products list upon successful completion of testing specified in table IV. Table IV is based on use of qualified MIL-S-8805/4 or MIL-S-8805/109 basic switches.



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TABLE IV. Qualification inspection - group submission.

| Inspection 1/ | Sample |
|---|---|
| Group I | All samples |
| Group II | M22885/75-201R (4 samples) |
| Group III | M22885/75-025W (2 samples) |
| Group VI 2/ | M22885/75-025W (2 samples) |
| Group VII | M22885/75-201R (2 samples) |
| Group VIII | 2 pushbuttons of each color (10 samples), of lens configuration 0, and without EMI/RFI shielding. 3/ |

1/ Inspection groups in accordance with qualification inspection table of MIL-S-22885.

2/ Electrical endurance: Test resistance load, ac, only. Lamps used during electrical endurance test shall be MS25237-387.

3/ Pushbuttons shall be assembled to any representative switch body.



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Part numbers: Part numbers are assigned as illustrated below. Part numbers do not include legend description. Acquire switches with legends as category II items in accordance with MIL-S-22885.

M22885/75 - 018 RG

- M22885/75: Specification sheet number
- 018: Dash number from table V
- RG: Color codes (from table III) for each lens segment, as required. Color code sequence is same as lens segment sequence shown on figure 2 below 1/

Example -

- 018: 3PDT momentary action switch with lens configuration 1 horizontal split), no EMI/RFI shielding, retained pushbutton.
- RG: Segment number 2 is red. Segment number 3 is green.

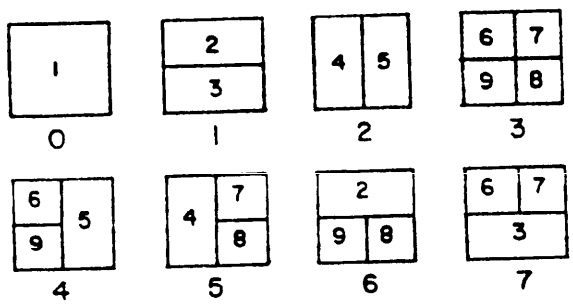



FIGURE 2. Lens configuration and segment numbers.

1/ Part number constructed as shown replace and are interchangeable with previous part numbers which included segment numbers in addition to color codes (example: M22885/75-018-2R3G).



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TABLE V. Dash numbers.

| Dash numbers | | Operation | Circuit | Lens configuration (see figure 2) | Retained pushbutton 1/ |
|---|---|---|---|---|---|
| Non-EMI shielded | EMI shielded | | | | |
| 001 | --- | Momentary | 1PDT | 0 | Yes |
| 002 | --- | " | " | 1 | " |
| 003 | --- | " | " | 2 | " |
| 2/ 004 | --- | " | " | 3 | " |
| 005 | --- | " | " | 4 | " |
| 006 | --- | " | " | 5 | " |
| 007 | --- | " | " | 6 | " |
| 008 | --- | " | " | 7 | " |
| 009 | 161 | Momentary | 2PDT | 0 | Yes |
| 010 | 162 | " | " | 1 | " |
| 011 | 163 | " | " | 2 | " |
| 012 | 164 | " | " | 3 | " |
| 013 | 165 | " | " | 4 | " |
| 014 | 166 | " | " | 5 | " |
| 015 | 167 | " | " | 6 | " |
| 016 | 168 | " | " | 7 | " |
| 017 | 169 | Momentary | 3PDT | 0 | Yes |
| 018 | 170 | " | " | 1 | " |
| 019 | 171 | " | " | 2 | " |
| 020 | 172 | " | " | 3 | " |
| 021 | 173 | " | " | 4 | " |
| 022 | 174 | " | " | 5 | " |
| 023 | 175 | " | " | 6 | " |
| 024 | 176 | " | " | 7 | " |
| 025 | 177 | Momentary | 4PDT | 0 | Yes |
| 026 | 178 | " | " | 1 | " |
| 027 | 179 | " | " | 2 | " |
| 028 | 180 | " | " | 3 | " |
| 029 | 181 | " | " | 4 | " |
| 030 | 182 | " | " | 5 | " |
| 031 | 183 | " | " | 6 | " |
| 032 | 184 | " | " | 7 | " |
| 033 | --- | Alternate | 1PDT | 0 | Yes |
| 034 | --- | " | " | 1 | " |
| 035 | --- | " | " | 2 | " |
| 2/ 036 | --- | " | " | 3 | " |
| 037 | --- | " | " | 4 | " |
| 038 | --- | " | " | 5 | " |
| 039 | --- | " | " | 6 | " |
| 040 | --- | " | " | 7 | " |
| 041 | 185 | Alternate | 2PDT | 0 | Yes |
| 042 | 186 | " | " | 1 | " |
| 043 | 187 | " | " | 2 | " |
| 044 | 188 | " | " | 3 | " |
| 045 | 189 | " | " | 4 | " |
| 046 | 190 | " | " | 5 | " |
| 047 | 191 | " | " | 6 | " |
| 048 | 192 | " | " | 7 | " |

See footnotes at end of table.



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TABLE V. Dash numbers - Continued.

| Dash numbers | | Operation | Circuit | Lens configuration (see figure 2) | Retained pushbutton 1/ |
|---|---|---|---|---|---|
| Non-EMI shielded | EMI shielded | | | | |
| 049 | 193 | Alternate | 3PDT | 0 | Yes |
| 050 | 194 | " | " | 1 | " |
| 051 | 195 | " | " | 2 | " |
| 052 | 196 | " | " | 3 | " |
| 053 | 197 | " | " | 4 | " |
| 054 | 198 | " | " | 5 | " |
| 055 | 199 | " | " | 6 | " |
| 056 | 200 | " | " | 7 | " |
| 057 | 201 | Alternate | 4PDT | 0 | Yes |
| 058 | 202 | " | " | 1 | " |
| 059 | 203 | " | " | 2 | " |
| 060 | 204 | " | " | 3 | " |
| 061 | 205 | " | " | 4 | " |
| 062 | 206 | " | " | 5 | " |
| 063 | 207 | " | " | 6 | " |
| 064 | 208 | " | " | 7 | " |
| 209 | 217 | Indicator | Indicator | 0 | Yes |
| 210 | 218 | " | " | 1 | " |
| 211 | 219 | " | " | 2 | " |
| 212 | 220 | " | " | 3 | " |
| 213 | 221 | " | " | 4 | " |
| 214 | 222 | " | " | 5 | " |
| 215 | 223 | " | " | 6 | " |
| 216 | 224 | " | " | 7 | " |
| 2/ 065 | 2/ 113 | Momentary | 2PDT | 0 | No |
| 066 | 114 | " | " | 1 | " |
| 067 | 115 | " | " | 2 | " |
| 068 | 116 | " | " | 3 | " |
| 069 | 117 | " | " | 4 | " |
| 070 | 118 | " | " | 5 | " |
| 071 | 119 | " | " | 6 | " |
| 072 | 120 | " | " | 7 | " |
| 2/ 073 | 2/ 121 | Momentary | 3PDT | 0 | No |
| 074 | 122 | " | " | 1 | " |
| 075 | 123 | " | " | 2 | " |
| 076 | 124 | " | " | 3 | " |
| 077 | 125 | " | " | 4 | " |
| 078 | 126 | " | " | 5 | " |
| 079 | 127 | " | " | 6 | " |
| 080 | 128 | " | " | 7 | " |
| 2/ 081 | 2/ 129 | Momentary | 4PDT | 0 | No |
| 082 | 130 | " | " | 1 | " |
| 083 | 131 | " | " | 2 | " |
| 084 | 132 | " | " | 3 | " |
| 085 | 133 | " | " | 4 | " |
| 086 | 134 | " | " | 5 | " |
| 087 | 135 | " | " | 6 | " |
| 088 | 136 | " | " | 7 | " |

See footnotes at end of table.



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TABLE V. Dash numbers - Continued.

| Dash numbers | | Operation | Circuit | Lens configuration (see figure 2) | Retained pushbutton 1/ |
|---|---|---|---|---|---|
| Non-EMI shielded | EMI shielded | | | | |
| 089 | 137 | Alternate | 2PDT | 0 | No |
| 090 | 138 | " | " | 1 | " |
| 091 | 139 | " | " | 2 | " |
| 2/ 092 | 2/ 140 | " | " | 3 | " |
| 093 | 141 | " | " | 4 | " |
| 094 | 142 | " | " | 5 | " |
| 095 | 143 | " | " | 6 | " |
| 096 | 144 | " | " | 7 | " |
| 097 | 145 | Alternate | 3PDT | 0 | No |
| 098 | 146 | " | " | 1 | " |
| 099 | 147 | " | " | 2 | " |
| 2/ 100 | 2/ 148 | " | " | 3 | " |
| 101 | 149 | " | " | 4 | " |
| 102 | 150 | " | " | 5 | " |
| 103 | 151 | " | " | 6 | " |
| 104 | 152 | " | " | 7 | " |
| 105 | 153 | Alternate | 4PDT | 0 | No |
| 106 | 154 | " | " | 1 | " |
| 107 | 155 | " | " | 2 | " |
| 2/ 108 | 2/ 156 | " | " | 3 | " |
| 109 | 157 | " | " | 4 | " |
| 110 | 158 | " | " | 5 | " |
| 111 | 159 | " | " | 6 | " |
| 112 | 160 | " | " | 7 | " |

1/ Retained pushbutton shall be capable of having lamps replaced but is restricted in complete removal from switch housing.

2/ Not for new design.

NOTE: 800 and 900 series dash numbers, for switch housing only and pushbutton only, have been canceled without replacement.



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## CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AV, MI
Navy - AS, OS
Air Force - 99
DLA - ES

User activities:
Army - AR
Navy - MC, SH
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DLA - ES

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